

Evaluation and regional differences analysis of the marine industry development level: The Case of China

Jianyue Ji^{a,b,1}, Huimin Liu^{a,1}, Xingmin Yin^{a,*,1}

^a School of Economics, Ocean University of China, Qingdao 266100, China

^b Institute of Marine Development, Ocean University of China, Qingdao 266100, China

ARTICLE INFO

Keywords:

Marine industry development
Comprehensive evaluation
Regional differences

ABSTRACT

The evaluation and regional differences analysis of marine industry development level play a significant role in the marine industry development of coastal regions. Taking China as an example, this study constructs a comprehensive evaluation system with three dimensions of innovation, industrial structure, and outcomes according to the connotation of marine industry development. The entropy-based TOPSIS method is applied to calculate the marine industry development level in 11 coastal provinces of China from 2007 to 2019, and the Dagum Gini coefficient is used to analyze regional differences. It is found that: (1) The weight ranking of the evaluation system is innovation, outcomes, and industrial structure in order, indicating that innovation can best reflect the marine industry development level. (2) The Chinese marine industry development level shows a positive trend, and the progress is especially evident after the strategy of building a strong maritime country is proposed. (3) The intra-regional and inter-regional differences are narrowing, indicating that the coordination of regional development is enhanced. (4) The primary source of regional differences is the intensity of trans-variation, so the government should pay attention to the coordinated development of marine industry among provinces, especially the backward provinces in the economic circle with high development level. The findings of this paper provide a useful reference for promoting China's marine industry development.

1. Introduction

As the world's population continues to grow and land resources are increasingly depleted, the oceans are attracting more and more attention from coastal countries [1,2]. The oceans have become a major area of international competition, and the marine economy is becoming a new growth point of the global economy [3,4]. Industry is an important part of the economy. The economy forms various industries through specialization and social division of labor, and the internal and external structure of the industries, in turn, determines the way of economic growth. The marine industry, which is an industrial sector that develops, utilizes, and protects marine resources and marine space, plays an important role in the development of the marine economy. Moreover, major marine countries in the world, such as the United States, the United Kingdom, and Japan, are adjusting their marine industry development strategies in line with the changes in the economy and

environment. It is foreseeable that promoting the integrated, coordinated, and sustainable development of the marine industry is the key task for coastal countries to strive for the competitive advantage of the marine economy for a long period in the future [5,6]. In the process of promoting the marine industry development, scientific evaluation of the marine industry development level is a fundamental and essential work, which can intuitively reveal its development trend and development law, and predict the future development direction, to provide support for the scientific decision-making of policymakers. However, there is still a lack of satisfactory methods to evaluate the marine industry development level.

China has a large population and vast ocean areas with a great abundance of marine resources [7]. As China's economy is increasingly dependent on marine resources and space, the marine industry is gradually becoming a new engine to drive its economic growth [8,9]. In 2012, the Chinese government formally put forward the strategy of

* Corresponding author.

E-mail address: yxm@ouc.edu.cn (X. Yin).

¹ Ocean University of China, 238, Songling Rd., Laoshan District, Qingdao, Shandong, China, Zip code: 266100, China

building a strong maritime country, and since then the development of the marine industry has been elevated to a new strategic height [10]. In recent years, the marine industry has made great progress in many aspects under the support of a series of policies, but there are still some issues that need to be addressed, such as the unreasonable industrial structure, the extensive growth mode of the marine industry, and unbalanced regional development level, which also are common challenges to many coastal countries [11–13]. Under these circumstances, in order to realize the healthy development of the marine industry, it is an important work to measure the development level of the marine industry and analyze its regional differences.

With the continuous improvement of the position of the ocean in human society, the development of the marine industry has gradually been paid attention to. In terms of the evaluation of the development level of the marine industry, the existing studies mainly adopted the single indicator evaluation method with the advantages of simplicity, easier access to data, and convenient calculation and the disadvantage of lacking rich connotation. For example, Shao et al. [14] and Wang et al. [15] interpreted marine industry development from the perspective of industrial structure upgrading, with the ratio of value added of the tertiary marine industry to the secondary marine industry and the ratio of value added of the secondary and tertiary marine industries to the value added of the primary marine industry respectively, while Xu et al. [16] illustrated the development level from the perspective of marine innovation capacity. Although such studies were valuable, it was difficult to fully illustrate the rich connotation of the marine industry development. Since comprehensive index evaluation can take into account various characteristics of the research object, it is gradually becoming a more prevalent evaluation approach when evaluating relatively complicated research objects. Halpern et al. [17, 18] developed and calculated the Ocean Health Index (OHI) to measure ocean health, and Anderson et al. [19] introduced the Fisheries Performance Index (FPI) to assess the performance of fisheries. However, there is a lack of studies on the comprehensive evaluation of the marine industry development level. The existing studies presented it as a part of the other evaluation system. For example, Yan et al. [10] used three indicators of gross product of marine primary, secondary, and tertiary industries to evaluate the marine industry development level when evaluating the competitiveness of the marine economy, while Song et al. [20] used the ratio of the value added of the marine tertiary industry to GOP and the ratio of the value added of the strategic emerging marine industry to GOP to evaluate the marine industry development when constructing the marine economic development index.

The existing studies have enriched the evaluation of the marine industry development level and provided valuable references for this study. However, there is a lack of comprehensive index evaluation studies focusing on the marine industry development level. Additionally, the analysis of regional differences also needs to be deepened. For these reasons, this study constructs a comprehensive index evaluation system with three dimensions of innovation, industrial structure, and outcomes to evaluate the marine industry development level and performs a regional difference analysis with the Dagum Gini coefficient.

The rest of the paper is organized as follows. Section 2 outlines the present status of China's marine industry development. Section 3 presents the evaluation of China's marine industry development level. Section 4 analyzes the regional differences in the marine industry development level. Lastly, conclusions and suggestions are presented in Section 5.

2. Present status of China's marine industry development

2.1. Progress

2.1.1. Keeping high growth in the marine industry

According to the marine industry division standard (GB/T20794–2006),¹ the marine industry consists of “the major marine industry” and “marine research, education, and management services”. As shown in Fig. 1, the value added of the marine industry increased from 1507.06 billion yuan to 5731.50 billion yuan from 2007 to 2019, with an average annual growth rate of 11.78%. The major marine industries generally maintain the growth trend, with the value added increasing from 1047.83 billion yuan to 3572.40 billion yuan, and an average annual growth rate of 10.76%. The marine research, education, and management service industries had always maintained growth, with the industry's value added increasing from 459.2 billion yuan to 2159.10 billion yuan, and an average annual growth rate peaking at 13.77%. Overall, from 2007 to 2019, the value added of China's marine industry maintained a year-on-year growth trend.

2.1.2. Continuous optimization in marine industry structure

With the improvement in the marine economy development level and the gradual accumulation of capital and technology, the marine secondary and tertiary industries have developed rapidly, and the structure of the marine industry has been optimized [21]. Before 1978, China's marine industries only included fisheries, marine salt industry, as well as marine communications and transportation industry. The total output value of the major marine industries was only a few billion yuan when the primary industry dominated, followed by the tertiary industry, and finally the secondary industry. Fig. 2 shows the changes in the output value of China's three marine industries from 2007 to 2019. During this period, gross ocean product (GOP) maintained a year-on-year growth trend, rising from 2561.87 billion yuan to 8941.60 billion yuan, an increase of 249.03%, and an average annual growth rate of 10.99%. The output value of the three marine industries continued to grow, where the value added of the primary, secondary, and tertiary industries increased from 139.54, 1201.10, and 1221.23 billion yuan to 372.90, 3198.70, and 5087 billion yuan, respectively, with an average annual growth rate of 8.54%, 8.50%, and 13.14%. The industrial pattern of “secondary and tertiary industries as the main industries, supplemented by primary industries” has basically been formed. Notably, the proportion of tertiary industries is increasing year by year.

2.1.3. Rapid development of the strategic emerging marine industries

As the country attaches importance to the ocean and national awareness of the ocean is improving, more funds and technologies are invested in the marine industry, and the ability to develop and utilize marine resources is increasing, the development of the strategic

¹ According to the marine industry classification standard (GB/T20794–2006): “The gross ocean product consists of two parts: marine industry and marine related industries. Marine industry is divided into major marine industry and marine scientific research, education, management and services, of which major marine industry is divided into 12 major industry categories, including marine fishery industry, offshore oil and natural gas industry, marine mining industry, sea salt industry, marine chemical industry, marine biomedical industry, marine electric power industry, marine seawater utilization industry, marine shipbuilding industry, marine engineering architecture industry, marine communication and transportation industry, coastal tourism.”

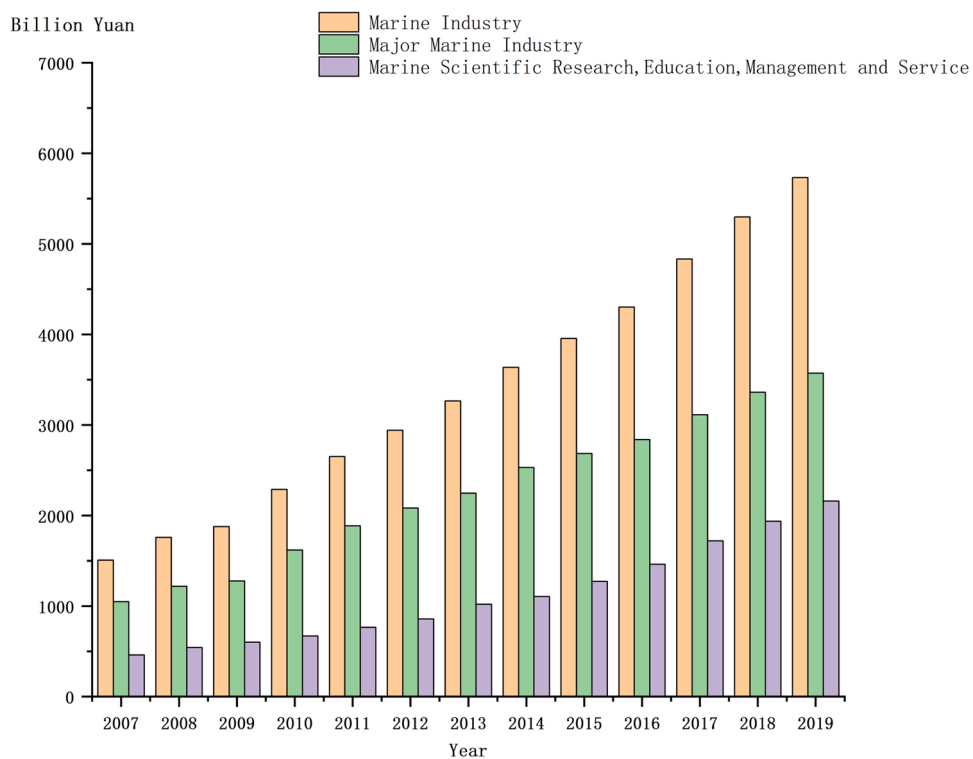


Fig. 1. Value added of marine industry products (2007–2019). Data sources: China Marine Statistical Yearbook (2008–2020).

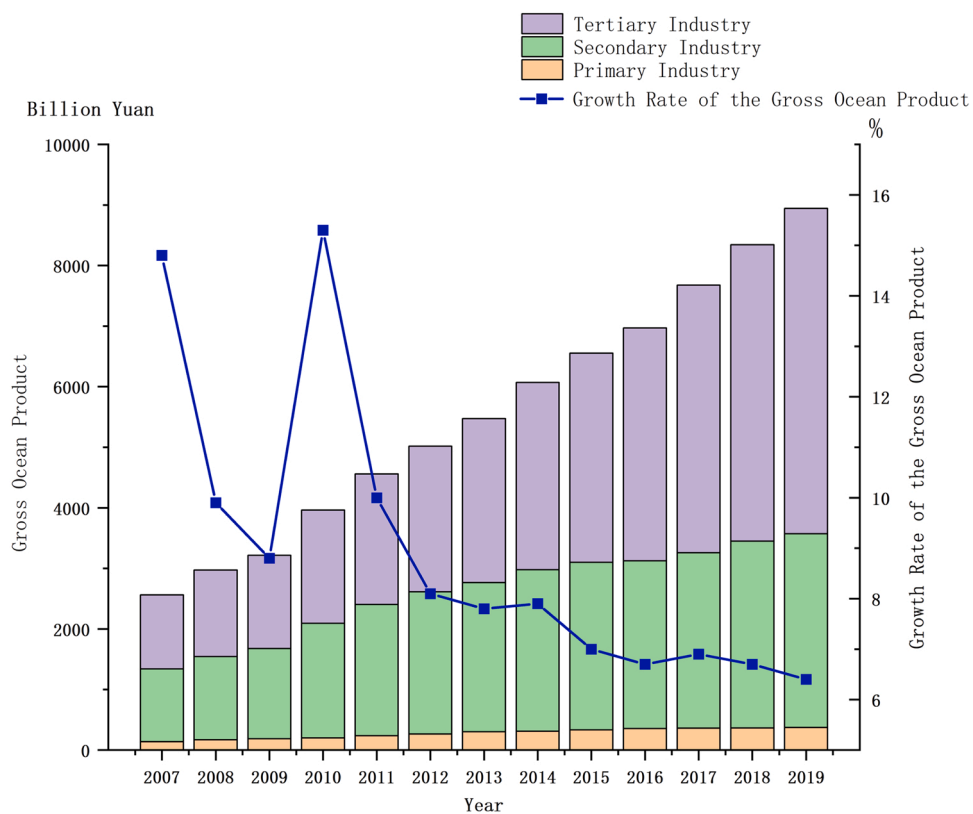


Fig. 2. GOP and its growth rate (2007–2019). Data sources: China Marine Statistical Yearbook (2008–2020).

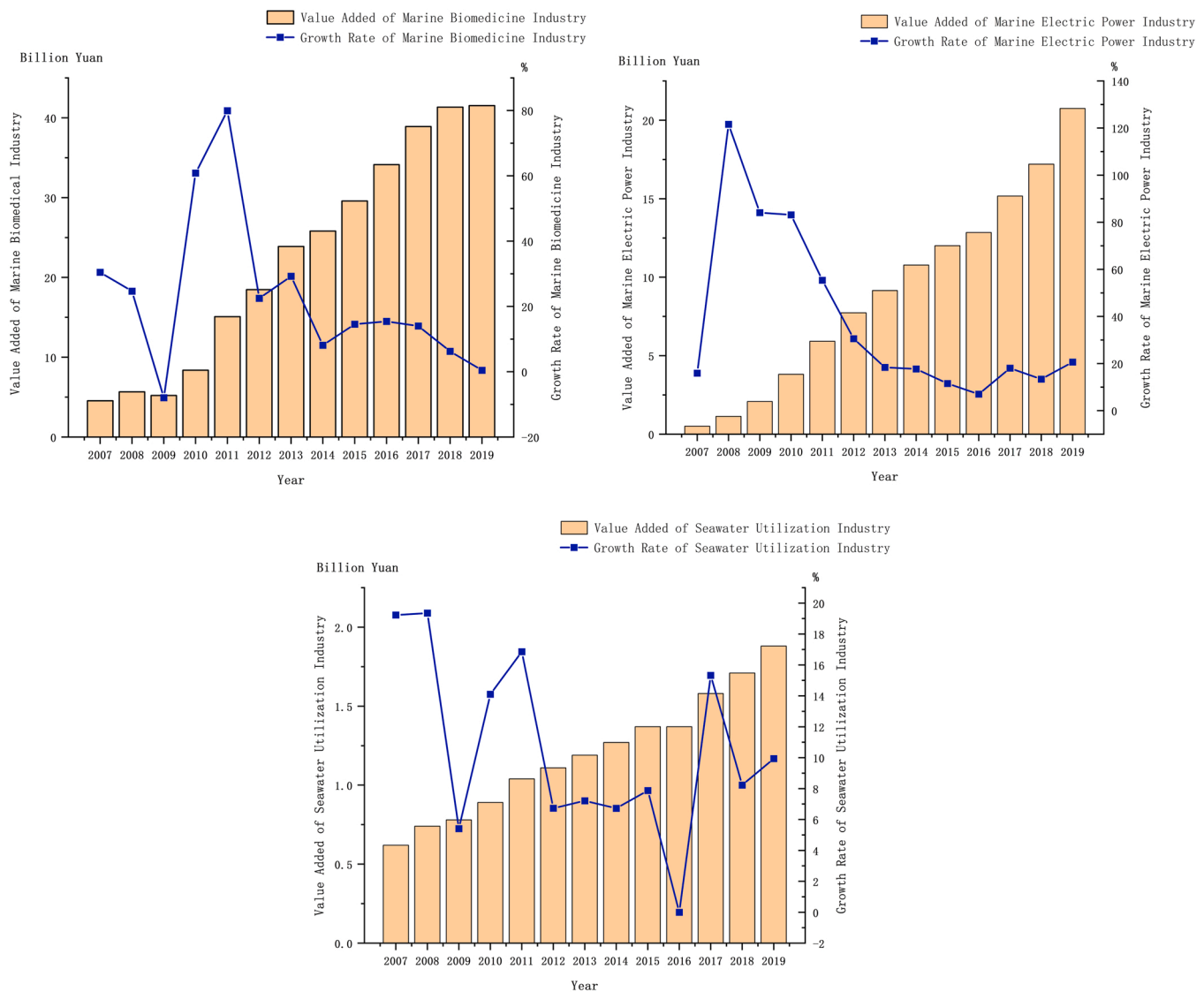


Fig. 3. The value added and growth rate of strategic emerging marine industries (2007–2019). Data sources: China Marine Statistical Yearbook (2008–2020).

emerging marine industries² is accelerating, and the marine industry system is gradually being completed [22]. During 2007–2019, breakthroughs made in the field of marine science and technology, and the strategic emerging marine industries achieved leapfrog developments, among which the average annual growth rates of the marine biomedical industry, marine electric power industry, and seawater utilization industry were 20.23%, 36.18%, and 9.68%, respectively (as shown in Fig. 3). The value added of the strategic emerging marine industries had achieved rapid growth, among which the marine electric power industry had the highest average yearly growth rate. Although the strategic emerging marine industries are developing rapidly, the scale is relatively small. Hence, the transformation of the strategic emerging marine industries into pillar industries should be actively facilitated.

² The National Development and Reform Commission pointed out in the "12th Five-Year Plan" development strategy study of strategic emerging marine industries published in March 2012 that: "the strategic emerging marine industries are industries supported by the development of marine high technology and large-scale development of marine resources, with a great leading role, strong industrial relevance, and great development potential, mainly including marine biomedical industry, marine electric power industry, marine seawater utilization industry, etc."

2.2. Problems

2.2.1. Unreasonable composition of major marine industries

Because the proportion of traditional marine industries is too high, the proportion of strategic emerging marine industries is relatively small, resulting in an unreasonable composition of the major marine industries. According to the relevant data from the statistical yearbook, the sum of the three traditional marine industries, namely coastal tourism, marine communications and transportation industry, and marine fishery, accounted for more than 80% of the major marine industry total value added, dominating the development of the entire major marine industry, while the remaining eight industries accounted for less than 20%. Fig. 4 reports the composition of the major marine industries' value added in 2019. The three traditional marine industries accounted for 84.38% of the major marine industries' total value added. However, the value added of emerging marine industries represented by the marine electric power industry, marine biomedicine, and seawater utilization industry accounted for 0.62%, 1.24%, and 0.06%, respectively. Thus, the developmental potential of emerging marine industries needs to be continuously explored.

2.2.2. The extensive mode of the marine industry development

For a long time, the marine industry has achieved extensive

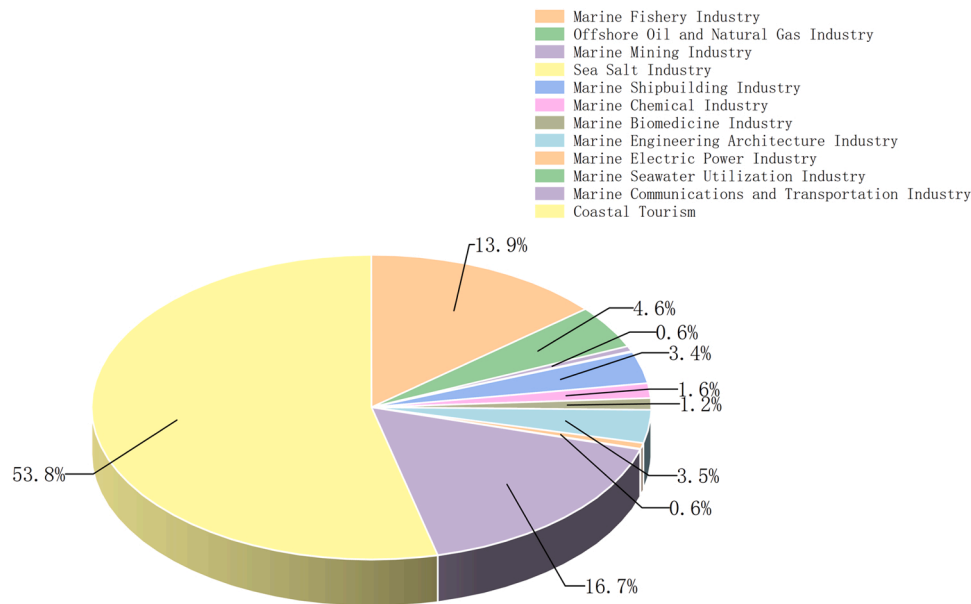


Fig. 4. The composition of the major marine industries value added (2019). Data sources: China Marine Statistical Yearbook (2020).

economic growth by increasing the input of production factors to expand the scale of production. Although this state has improved significantly in recent years, the problem of extensive marine industry development remains very prominent. Take the marine mining industry as an example, China's coastal mineral resources are rich, and more than 60 kinds of minerals have been discovered, but only 10 kinds of minerals have been currently mined on a large scale, mainly because of the relatively backward resource exploration and mining technology. In terms of resource exploration, equipment is not advanced which makes it impossible to conduct a comprehensive exploration of mineral resources, resulting in a low recovery rate of minerals and a serious waste of resources. In terms of resource extraction, the mining process is mostly in an extensive stage, with mining and beneficiation technology generally at a low level, resulting in only one or a few minerals being mined. Hence, many useful minerals are abandoned, which also causes much waste of minerals. In addition, mining activities and their waste emissions also occupy land resources, further aggravating the consumption of resources. Therefore, to achieve further development of the marine industry, the development model of high input and high energy consumption needs to be changed, and the economic efficiency of the marine industry should be improved by upgrading the technology level to increase output and realize the change from externally-oriented economic growth to internally oriented economic growth.

2.2.3. Environmental pollution problems induced by the marine industry

Owing to the excessive exploitation of marine resources, the marine environment has been negatively affected to a certain extent [23]. Take the mariculture industry as an example. The bait and various chemical drugs in the mariculture process cannot be fully used by aquatic organisms, resulting in eutrophication and accumulation of toxic substances in the water body of the breeding area, which in turn induces offshore water pollution, red tide, marsh moss flooding, and other frequent marine disasters. This has caused marine ecosystem degradation. For example, the 2018 survey on the health status of China's marine ecosystems shows that in the whole monitoring system, the marine ecosystems in healthy, sub-healthy, and unhealthy environments were 23.8%, 71.4%, and 4.8%, respectively, and the health status of marine ecosystems was not optimistic.

On the other hand, marine pollution has also caused severe economic losses. According to China Marine Statistical Yearbook 2013, red tide disasters occurred 73 times, covering a total area of 7971 square

kilometers, causing direct economic losses of 2.015 billion yuan in 2012. Therefore, green, coordinated, and sustainable development should also be the main direction for the future development of the marine industry.

In summary, there are positive and negative aspects of the marine industry's development. It is impossible to directly determine whether the marine industry development level is increasing or decreasing through statistical data. And it is even more difficult to make an accurate and reasonable comparison among different regions, which requires us to adopt a scientific approach for comprehensive evaluation. What is the changing trend in the development level of China's marine industry? What are the differences in the development level of the marine industry in different regions? The answers to the above questions cannot only observe and grasp the real state of China's marine industry but also provide a new perspective for understanding realistic marine economic development in China. Therefore, the first question to be addressed in this study is the evaluation of the marine industry development level.

3. Evaluation of China's marine industry development level

3.1. The connotation of the marine industry development

According to Schumpeter's view [24], the purpose of economic activities is to satisfy human needs. As an important economic activity, the goal of marine industry development is also to meet human needs which can be measured by the increase in desirable output and the decrease in undesirable output of the marine industry. To achieve this goal, it must rely on innovation, which is the driving force of industrial development. Under the impetus of innovation, the sectors in the marine industry follow the development direction of green, advanced, and efficient, realizing the reallocation of production factors in each sector, and then leading to changes in the marine industry structure. On this basis, the outcomes of marine industry development are manifested in the improvement of economic and environmental benefits [25]. In summary, marine industry development can be summarized as the process of the marine industry taking innovation as the impetus to promote the continuous optimization of marine industry structure, enhancing the economic and environmental benefits to meet human needs of economic and social development. Fig. 5 graphically shows the process of the marine industry development.

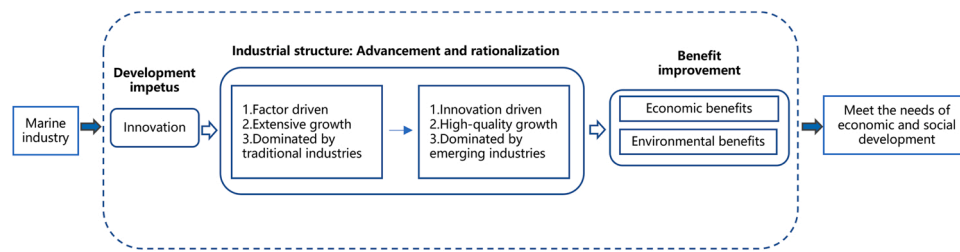


Fig. 5. The process of the marine industry development.

3.2. Establishment of index system

According to the connotation of marine industry development, this study constructs a comprehensive evaluation system from three dimensions of innovation, industrial structure, and outcomes.

Innovation: Innovation is not only the driving force of an industry's development but also reflects its prospects and future, the United States, Japan, and other countries with strong marine industries also continue to launch relevant policies to support the progress of marine science and technology [26,27]. Thus, innovation ability is an essential factor in measuring the marine industry development.

Because a huge investment is needed for innovation, funding investment and talent investment are regarded as significant indicators of innovation capability in the marine industry. In terms of funding, the marine R&D expenditure and intensity of marine R&D expenditure (the proportion of marine R&D expenditure to GOP) selected in this study can represent the total amount of funds invested in marine science and technology innovation and the intensity of investment in each coastal province [26]. In addition, due to the shortage of marine research talents, very few marine-related enterprises have independent product R&D departments, and most of the technological innovation is realized through scientific research institutions or joint university-enterprise cooperation. This means that the number of marine research personnel in marine research institutions and the proportion of marine research talents with a bachelor's degree or above selected in this study can reflect the region's quantity and quality of marine research talents.

Notably, even though the number of invention patents is often regarded as an effective indicator of innovation capability, many scholars point out that some enterprises have applied for non-utility patents that do not reflect the innovative ability of enterprises in order to obtain government subsidies in recent years [27]. Furthermore, the annual fluctuation in the number of patents granted in each region is large, which may be highly related to the authorization standards and relevant policies in different years and regions, then has led to the discrepancy between their innovation level and the actual situation. Therefore, the patent indicator is not incorporated into the index system in this study.

Industrial structure: Advancement and rationalization are used in this study to evaluate the industrial structure of the marine industry [28].

Advancement is essentially the transformation of product value added from low to high. From the perspective of the three industries, advancement is the process that the focus of the industrial structure gradually shifts from primary industry to secondary industry and tertiary industry. From the perspective of factor intensity, it shifts from labor-intensive to capital-, technology-, and knowledge-intensive [29]. Hence, industrial structure upgrading not only covers the change in the proportion of the three industries but also includes the optimization of the technology level and the change in the leading factors within the industry. Therefore, this study draws on previous research to construct an industrial structure upgrading index that reflects optimization among and within industries.

Moreover, this study found that the percentage of value added of emerging marine industries, a commonly used index to evaluate

industrial structure advancement, is not available based on existing data. However, the statistical index of "value added of marine research, education and management services" includes the value added of marine information services, marine technical services, marine scientific research, marine management, and other scientific research, education, management, and service activities in the process of developing, utilizing, and protecting the ocean. Accordingly, the greater the proportion of the value added of marine research, education, and management services to the value added of GOP, the more it can reflect the development of the marine industry in the direction of capital-intensive, technology-intensive, and knowledge-intensive. Therefore, this study considered the ratio of the marine research, education, and management services value added to the GOP value added as another indicator to evaluate the advancement of industrial structure.

On the other hand, rationalization focuses more on coordination among industries and the effective use of resources [28]. The absence of statistics on the number of people employed in the three marine industries makes it impossible to obtain indicators such as the Thiel index, the industrial structure deviation index, and the Krugman coefficient of variation of industrial structure, which are usually used to evaluate the rationalization of industrial structure [30–32]. For this reason, this study innovated the measurement method of rationalization based on the existing statistical data and the essence of rationalization and then constructed the marine industry coupling and coordination index, and the marine industry structure Moore index, whose construction principles were as follows.

- (1) Marine industry coupling coordination index: In terms of the characteristics of the marine industry itself, the major marine industries and marine related industries complement each other. The higher the degree of coupling and coordination between the two, and the more complete the industrial chain is, the more conducive to the upgrading of the marine industry structure. Therefore, this study naturally associates the construction of a coupling coordination index between major marine industries and marine-related industries [33]. The higher the degree of coupling and coordination between them, the more reasonable the structure of the marine industry.
- (2) Marine industry structure Moore index: The more drastic changes in the industrial structure on the premise that the production factors can flow freely, represents a more rational industrial structure as the production factors become more and more coordinated in proportion among the sectors and more efficient in resource allocation under the action of the market. Thus, this study constructed a marine industrial structure based on the connotation of the Moore index [34]. Essentially, the larger the Moore index of the marine industrial structure is, the more drastic the changes in the structure of the three industries, the more efficient the allocation of resources of production factors in each sector, and the more reasonable the industrial structure.

Outcomes: The goal of industrial development is to better meet human needs. Therefore, the benefits of the marine industry should be emphasized in the evaluation of the marine industry development level.

Ideally, we hope that the marine industry development can move towards high quality and sustainability, and achieve a balance between economic and environmental benefits [29,35].

In terms of economic benefits, this study selected three indicators: GOP, the ratio of GOP to GDP, and labor output rate to characterize the total amount of marine economy in each coastal area, the contribution of the marine industry to the local economy, and the productivity of the marine industry, respectively. The higher the three indicators, the higher the economic benefits of the marine industry.

In terms of environmental benefits, the offshore water quality status and wastewater emissions per unit of GOP were selected to reflect the environmental benefits, where better offshore water quality and lower wastewater emissions per unit of GOP can reflect the marine industry development in green, low-carbon, and sustainable directions.

Accordingly, this study considered innovation, industrial structure, and outcomes as the core, and the representativeness, reasonableness, and accessibility of evaluation indexes as the primary screening principles. Finally, a marine industry development evaluation index system was constructed, including three subsystems and six guideline layers with a total of 13 evaluation indicators (Table 1).

This study takes 11 coastal provinces of China (excluding Hong Kong, Macao, and Taiwan for the sake of data availability) from 2007 to 2019 as samples, and the data are from the China Marine Economic Statistical Yearbook, the China Statistical Yearbook, and the China Offshore Marine Environment Quality Bulletin. In the evaluation, this paper takes 2006 as the base year and uses the GDP Deflator to eliminate the influence of inflation on the relevant indicators. Most indicators can be obtained directly or through simple calculation, yet the Marine Industry Coupling Coordination Index, the Marine Industry Structure Moore Index, and the Marine Industry Structure Upgrading Index are all derived through more complex processing. Here are the specifics:

- (1) Marine industry coupling coordination index: Drawing on other researches [36,37], this study constructs the marine industry coupling coordination index, which is calculated by the following formulas:

$$C = \frac{2\sqrt{u_1 u_2}}{u_1 + u_2} \quad (1)$$

$$D = \sqrt{C \cdot T}, T = \alpha u_1 + \beta u_2 \quad (2)$$

where u_1 and u_2 represent the growth rate of value added of the marine industry and value added of marine-related industries, respectively. C is the coupling degree between them, the closer it is to 1, the higher the coupling degree; D represents the coupling degree; T is the comprehensive coordination index reflecting marine industry and marine-related industries, where α and β are

taken as 0.5.

- (2) Marine industry structure Moore index: This indicator is designed in conjunction with related research [34], and the calculation formula is as follows:

$$MSMI_i = \frac{\sum_{s=1}^3 (industry_{si,2006} \times industry_{si,r})}{\sqrt{\sum_{s=1}^3 (industry_{si,2006})^2 \times \sum_{s=1}^3 (industry_{si,r})^2}} \quad (3)$$

where $industry_{si,r}$ represents the share of industries in region i of year r , $s = 1, 2, 3$.

- (3) Marine structure upgrading index: This indicator is based on relevant literature [38] and designed as follows:

$$MSUI_i = \sum_{s=1}^3 industry_{si,r} \times s \quad (4)$$

3.3. Calculation of the marine industry development index based on entropy-based TOPSIS

The entropy-weighted TOPSIS method is a popular evaluation method for performing comprehensive evaluations. Yan et al. [10] used it to evaluate the sustainability level of China's marine economy, and V. Salehi et al. [39] used it to analyze and evaluate crisis management systems in the petrochemical industry. Drawing on previous studies [40–42], the entropy-based TOPSIS method which is essentially an improvement over the traditional TOPSIS is applied to calculate the marine industry development index (hereafter this text will be abbreviated as MIDI). The key of it is to define the distance between the optimal and inferior solutions of the decision problem, then calculate the relative distance of each situation to an ideal solution, and finally rank the superiority and inferiority. This method can meet the need for scientific ranking of evaluation results, but it cannot reflect the significance of different indicators in an index system. However, the introduction of the entropy method can solve this problem as well as exclude the interference of anthropic factors. In addition, to compare different years in each region, the time variable is added to the entropy-based TOPSIS model, and the specific evaluation model is as follows:

- (1) Indicator setting: X_{rij} represents the value of the indicator j of province i in year r , $j = 1, 2, 3, \dots, M$; $i = 1, 2, 3, \dots, N$; $r = 1, 2, 3, \dots, T$.

- (2) Standardization of indicators: This study applies the maximum difference normalization method to standardize each indicator X_{rij} in the marine industry development evaluation system to eliminate the differences in the indexes' order of magnitude and magnitude:

Table 1
Marine industry development evaluation index system.

Goals	Subsystems	Guideline	Specific evaluation indicators	Unit	Properties
Marine industry development level	Innovation	Funding	Marine R&D expenditures	Thousand	+
			Marine R&D spending strength	%	+
	Industry Structure	Talent	Number of marine researchers	People	+
			The ratio of undergraduate and above marine research talents	%	+
		Rationalization	Marine industry coupling coordination index	—	+
			Marine industry structure Moore index	—	+
		Advancement	Marine industry structure upgrading index	—	+
			The proportion of marine research and education management services	%	+
	Outcomes	Economic benefits	GOP	Billion	+
			Share of GOP	%	+
			Marine industry labor output rate	RMB 10,000/person	+
		Environmental benefits	Offshore water quality status	%	+
			Wastewater emissions per unit of GOP	Ton/RMB10,000	—

$$Y_{rij} = \begin{cases} \frac{X_{rij} - X_{\min}}{X_{\max} - X_{\min}}, & X_{rij} \text{ is a positive indicator} \\ \frac{X_{\max} - X_{rij}}{X_{\max} - X_{\min}}, & X_{rij} \text{ is a negative indicator} \end{cases} \quad (5)$$

where X_{rij} and Y_{rij} represent the original and standardized marine industry development evaluation indicator values, respectively, and $X_{\max}$ and $X_{\min}$ represent the maximum and minimum values of X_{rij} , respectively.

(3) The information entropy E_j of each standardized Y_{rij} in the index system of the marine industry development level is calculated.

$$E_j = -k \sum_{r=1}^T \sum_{i=1}^N Z_{rij} \ln(Z_{rij}) \quad (6)$$

where $k > 0, k = 1/\ln(TN)$, $Z_{rij} = Y_{rij} / \sum_{r=1}^T \sum_{i=1}^N Y_{rij}$

(4) Calculate the information utility value of the j evaluation indicator:

$$G_j = 1 - E_j \quad (7)$$

(5) Calculate the weights of the j evaluation indicator:

$$W_j = \frac{G_j}{\sum_{j=1}^M G_j} \quad (8)$$

(6) Construct the weighting matrix H_{ri} for each year of evaluation indicator:

$$H_{ri} = (h_{ri})_{TN \times M} \quad (9)$$

where $h_{ri} = W_j \times Y_{rij}$.

(7) Determine the Euclidean distances d_{ri}^+ and d_{ri}^- for the optimal solution Q_j^+ and the inferior solution Q_j^- from the weighting matrix H_{ri} .

$$d_{ri}^+ = \sqrt{\sum_{j=1}^M (Q_j^+ - h_{rij})^2} \quad (10)$$

$$d_{ri}^- = \sqrt{\sum_{j=1}^M (Q_j^- - h_{rij})^2} \quad (11)$$

(8) Calculate the $MIDI_{ri}$:

$$MIDI_{ri} = \frac{d_{ri}^-}{d_{ri}^- + d_{ri}^+} \times 100 \quad (12)$$

where $MIDI_{ri}$ is between 0 and 100; the larger $MIDI_{ri}$ is, the better the marine industry development of province i in year r is; otherwise, the worse the situation.

3.4. The results of the marine industry development level

3.4.1. Weights

Table 2 shows the weights of every indicator, the innovation subsystem has the largest weight, reaching 0.44, indicating that the innovation subsystem has the most noticeable changes during the observation period and provides the most useful information. The second weight is the outcomes subsystem, which reaches 0.38, indicating that benefits improvement is also important in the development of the marine industry. Finally, with a weight of only 0.18, the industrial structure subsystem indicated that the variation in the data in the industrial structure subsystem is small and provides less information. In other words, the industrial structure is more stable than the innovation power and comprehensive efficiency during the observation period, which is also consistent with the real situation.

3.4.2. The whole country

From the national level (Fig. 6), the MIDI showed a rising trend year by year and had an obvious multistage. From 2008–2012, the MIDI was in a slow upgrading stage, and the MIDI rose gradually from 23.96 to 27.77, probably because of the global financial crisis, the marine industry also suffered a certain impact, and the growth rate of the MIDI slowed down; 2012–2015 was a period of rapid development of the marine industry. The MIDI increased rapidly from 27.77 to 34.56. At that time, the negative impact of the financial crisis receded, and the country paid more and more attention to the development of the marine economy in the meantime. In particular, the China government proposed a strategy for building a strong maritime country for the first time, and then a series of related policies were launched one after another to promote the development of the marine industry. During 2015–2019, the MIDI rose immediately after a slight rebound, probably because China started to carry out supply-side reform in 2015, during which some marine industries with excess capacity were de-capacitated, which caused a short-term decline in economic benefits. However, in the long term, it was conducive to the optimization of industrial restructuring and further improvement of comprehensive benefits. The index dropped from 34.56 to 33.54 in 2016, then rose to 34.48 in 2017. In 2018, China government proposed to accelerate the establishment of a strong maritime country and further promote the development of the marine industry. Consequently, the marine industry development index rose rapidly to 41.92.

3.4.3. Marine economic circles

The overall situation of the three marine economic circles³ was on an upward trend, and the trend coincides with the national trend and showed an obvious multistage. As shown in Fig. 7, the ranking of MIDI in the three marine economic circles was East > North > South, showing the characteristics of "high in the east, middle in the north, and low in the south." The MIDI in the eastern marine economic circle was significantly higher than the national average, but the advantage gradually decreased over time. The northern marine economic circle was roughly similar to the national average, and the southern marine economic circle was lower than the national overall, but the gap continued to narrow. From the time evolution trend, the curves of the three marine economic circles were almost parallel before 2015 when the regional gaps among the marine industry development level of the three major marine economic circles were apparent. Then, the gaps among those kept narrowing after 2015. Until 2018, the gaps among those were small.

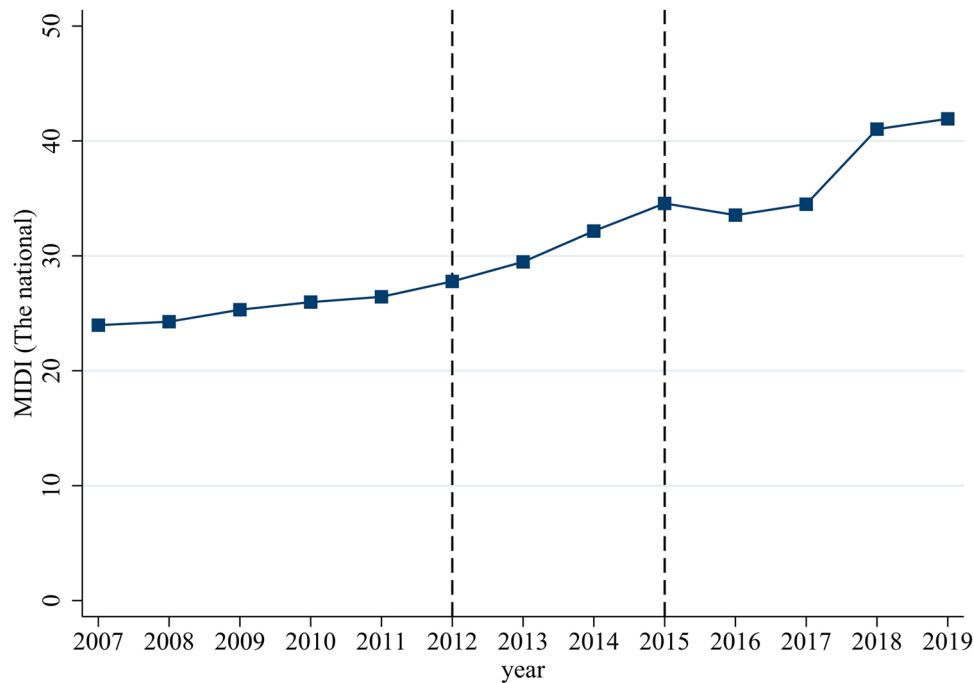
3.4.4. Coastal provinces

Table 3 reports the MIDI in coastal provinces. According to the results, this study found that: (1) MIDI at the end of the observation period was higher than at the beginning of the period, but the increase varies considerably from province to province. (2) From the average value of all years, there were large gaps among different provinces, and only four provinces had an average value that was higher than the national average, namely, Shanghai, Shandong, Guangdong, and Tianjin. Among them, the three provinces that always ranked high were Shanghai, Shandong, and Guangdong, and the ranking of the three provinces changed from "Shanghai-Shandong-Guangdong" at the beginning of the period to "Guangdong-Shandong-Shanghai" at the end of the period, while Tianjin was overtaken by Liaoning and Jiangsu in the later period.

³ According to the outline of the 14th Five-Year Plan, the three major marine economic circles are the northern marine economic circle, the eastern marine economic circle, and the southern marine economic circle. The northern marine economic circle includes Liaoning Province, Hebei Province, Tianjin City, and Shandong Province. The eastern marine economic circle includes Jiangsu Province, Shanghai City, and Zhejiang Province. The southern marine economic circle includes Fujian Province, Guangdong Province, Guangxi Zhuang Autonomous Region, and Hainan Province.

Table 2
Weights.

Subsystems	Guideline	Specific evaluation indicators	Weights			Rank
Innovation	Funding	Marine R&D expenditures	0.4597	0.3163	0.1949	1
		Marine R&D spending strength			0.1213	4
	Talent	Number of marine researchers		0.1434	0.1214	3
Industry Structure	Rationalization	The ratio of undergraduate and above marine research talents	0.1460	0.0144	0.0220	10
		Marine industry coupling coordination index			0.0052	13
		Marine industry structure Moore index			0.0092	12
	Advancement	Marine industry structure upgrading index		0.1316	0.0445	8
		The proportion of marine research and education management services			0.0871	7
Outcomes	Economic benefits	GDP	0.3943	0.3424	0.1376	2
		Share of GDP			0.1067	5
		Marine industry labor output rate			0.0981	6
	Environmental benefits	Offshore water quality status		0.0519	0.0325	9
		Wastewater emissions per unit of GDP			0.0195	11

**Fig. 6.** MIDI in the whole country (2007–2019).

(3) From the fluctuation of the MIDI during the observation period, the fluctuation of each province before 2013 was relatively small, and after 2013, the overall trend of more obvious upward fluctuation appears.

4. Analysis of regional differences of marine industry development level based on the Dagum Gini coefficient

To further reveal the extent of regional differences and sources of differences in the development level of China's marine industry, this study refers to relevant studies [43] and uses the Dagum Gini coefficient and its decomposition to evaluate the overall differences, intra-regional differences, inter-regional differences, and contribution rates of the marine industry development level.

The Dagum Gini coefficient is adopted to evaluate the intra-regional (intra-group) and inter-regional (inter-group) differences among China's three marine economic circles and measures the contribution rates of differences' sources. Generally, the smaller the intra-regional Gini coefficient, the closer the development level of the marine industry is among the provinces in the group. The larger the intra-regional Gini coefficient, the larger the gaps among the development level of the marine industry in the provinces within the group; the inter-regional and intra-regional gaps are similar. The total Gini G can be decomposed into the sum of the intra-regional difference contribution G_w ,

inter-regional difference contribution G_{nb} and intensity of transvariation contribution G_l , that is, $G = G_w + G_{nb} + G_l$. The specific calculation formulas are as follows:

$$G = \frac{\left(\sum_{i=1}^k \sum_{m=1}^k \sum_{j=1}^{h_i} \sum_{n=1}^{h_j} |y_{im} - y_{jn}| \right)}{2h^2\bar{y}} \quad (13)$$

$$G_{ii} = \frac{\frac{1}{2\bar{y}_i} \sum_{m=1}^{h_i} \sum_{n=1}^{h_i} |y_{im} - y_{jn}|}{h^2} \quad (14)$$

$$G_w = \sum_{i=1}^k G_{ii} P_i S_i \quad (15)$$

$$G_{ij} = \frac{\sum_{m=1}^{h_i} \sum_{n=1}^{h_j} |y_{im} - y_{jn}|}{h_i h_j (\bar{y}_i - \bar{y}_j)} \quad (16)$$

$$G_{nb} = \sum_{i=2}^k \sum_{j=1}^{i-1} G_{ij} (p_i s_j + p_j s_i) D_{ij} \quad (17)$$

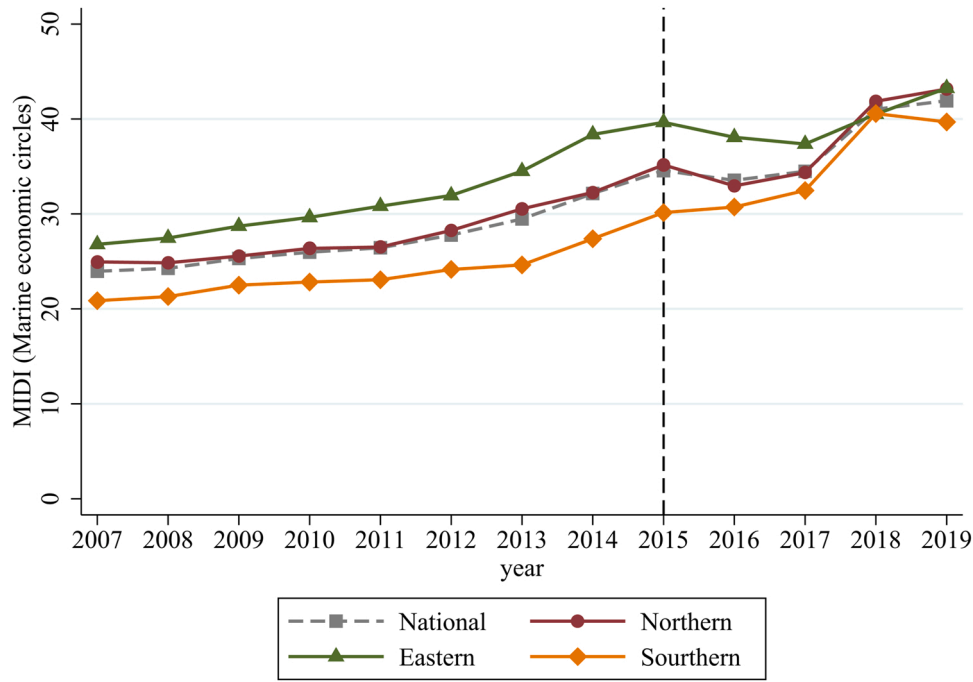


Fig. 7. MIDI in marine economic circles (2007–2019).

Table 3
Marine industry development level.

Year	Nationwide	Northern Marine Economic Circle				Eastern Marine Economic Circle			Southern Marine Economic Circle			
		Tian jin	He bei	Liao ning	Shan dong	Shanghai	Jiang su	Zhe jiang	Fu jian	Guangdong	Guang xi	Hai nan
2007	23.96	32.94	14.11	16.25	36.48	44.79	17.14	18.45	21.89	27.48	10.98	23.10
2008	24.27	30.98	15.31	16.91	36.21	46.22	18.00	18.17	22.42	28.39	11.67	22.73
2009	25.31	31.95	12.62	18.60	39.08	46.77	19.89	19.50	23.56	32.28	11.31	22.85
2010	25.97	32.75	12.85	19.60	40.28	47.95	22.18	18.79	23.32	34.16	11.48	22.35
2011	26.44	31.34	13.44	20.20	41.08	46.57	26.10	19.80	23.34	35.78	10.72	22.44
2012	27.77	33.03	14.33	21.49	44.20	49.28	25.65	20.91	23.04	39.21	11.79	22.58
2013	29.47	34.72	14.97	25.57	46.89	54.28	27.73	21.50	23.45	41.11	11.02	22.96
2014	32.16	36.81	16.84	25.44	49.95	58.71	33.31	23.11	25.99	47.11	13.76	22.69
2015	34.56	38.13	17.71	35.57	49.26	61.54	31.42	25.99	28.93	54.28	13.72	23.63
2016	33.54	33.74	16.66	35.61	45.82	57.12	30.78	26.33	30.51	55.54	12.65	24.22
2017	34.49	33.82	18.71	36.51	48.38	52.47	31.56	28.06	32.44	60.21	12.52	24.70
2018	41.02	33.68	27.67	41.77	64.33	57.67	34.16	29.69	33.55	70.48	22.58	35.67
2019	41.92	37.80	27.48	45.09	62.27	62.78	34.71	32.25	33.76	71.99	17.45	35.55
Mean	30.80	33.98	17.13	27.59	46.48	52.78	27.12	23.27	26.63	46.00	13.20	25.04
Rank	—	4	10	5	2	1	6	9	7	3	11	8

$$G_i = \sum_{i=2}^k \sum_{j=1}^{i-1} G_{ij} (p_i s_j + p_j s_i) (1 - D_{ij}) \quad (18)$$

$$D_{ij} = \frac{(d_{ij} - p_{ij})}{(d_{ij} + p_{ij})} \quad (19)$$

$$d_{ij} = \int_0^{\infty} dF_i(y) \int_0^y (y-x) dF_j(x) \quad (20)$$

$$p_{ij} = \int_0^{\infty} dF_j(y) \int_0^y (y-x) dF_i(x) \quad (21)$$

where i and j are different marine economic circles; m and n are different coastal provinces; k is the number of marine economic circles; h_i (h_j)

indicates the number of coastal provinces in $i(j)$, $y_{im}(y_{jn})$ is the MIDI in coastal provinces $m(n)$ within marine economic circles $i(j)$, and $\bar{y}$ indicates the average value of the MIDI in all provinces. Eqs. (14) and (15) represent the intra-regional Gini coefficient in marine economic circle i and the intra-regional difference contribution in all coastal provinces, respectively; Eqs. (16) and (17) represent the inter-regional Gini coefficient between i and j and the inter-regional net difference contribution in all provinces, respectively. (18) represents the contribution of the intensity of transvariation, D_{ij} is the relative impact of marine industry development between region i and region j , and its calculation equation is shown in Eq. (19); d_{ij} and p_{ij} represent the mathematical expectation of the sum of all sample values in region i and region j that are consistent with $y_{im} - y_{jn} > 0$ and $y_{jn} - y_{im} > 0$, and their calculation equations are shown in Eqs. (20) and (21), respectively. $F_i(F_j)$ represents the cumulative density distribution function in region $i(j)$.

As shown in Table 4, column (1) showed the total Gini coefficient of the marine industry development level with a mean value of 0.2266,

Table 4
Dagum Gini coefficient.

Year	Total	intra-regional differences			inter-regional differences			Contribution rate (%)		
		northern marine economic circle	eastern marine economic circle	southern marine economic circle	northern – eastern	northern – southern	eastern – southern	intra-regional	inter-regional	intensity of transvariation
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
2007	0.2304	0.2100	0.2293	0.1519	0.2537	0.2361	0.2580	28.4195	23.7707	47.8098
2008	0.2206	0.1931	0.2284	0.1481	0.2476	0.2135	0.2582	28.3535	25.0236	46.6229
2009	0.2304	0.2267	0.2111	0.1767	0.2525	0.2303	0.2491	29.8929	22.8944	47.2127
2010	0.2327	0.2262	0.2186	0.1890	0.2516	0.2339	0.2466	30.4649	24.2918	45.2433
2011	0.2274	0.2218	0.1930	0.2062	0.2305	0.2375	0.2424	30.7522	27.4735	41.7743
2012	0.2309	0.2237	0.1973	0.2141	0.2326	0.2415	0.2448	31.0082	26.2171	42.7747
2013	0.2415	0.2147	0.2111	0.2303	0.2373	0.2532	0.2695	30.3310	30.2608	39.4081
2014	0.2394	0.2145	0.2062	0.2358	0.2353	0.2447	0.2702	30.6240	30.4090	38.9669
2015	0.2358	0.1728	0.1993	0.2633	0.2241	0.2561	0.2654	29.9373	25.1667	44.8961
2016	0.2289	0.1694	0.1797	0.2745	0.2106	0.2543	0.2476	30.7755	19.9382	49.2863
2017	0.2222	0.1668	0.1452	0.2903	0.1780	0.2586	0.2462	31.4160	13.4826	55.1014
2018	0.1968	0.1763	0.1535	0.2247	0.1761	0.2134	0.2074	32.7370	3.8642	63.3988
2019	0.2089	0.1617	0.1569	0.2605	0.1735	0.2376	0.2276	31.9170	9.3686	58.7144
means	0.2266	0.1983	0.1946	0.2204	0.2233	0.2393	0.2487	30.5099	21.7047	47.7854

which showed that the spatial non-equilibrium of the marine industry development level was more significant. In addition, the total Gini coefficient decreased from 0.2304 to 0.2089 during the observation period, with an average annual decrease of 0.81%, indicating that the total gap in China's marine industry development level was narrowing. According to the evolutionary process, the total Gini coefficient showed a fluctuating upward trend until 2013, and the total Gini coefficient reached its maximum in 2013 and generally had been in decline thereafter.

Columns (2) to (4) show the evolution of intra-regional differences in the three marine economic circles. The intra-regional differences in the northern and eastern marine economic circles tended to generally decline, with an average annual decreasing rate of 2.15% and 3.12%, respectively, which showed an increase in the coordination of intra-regional development. Meanwhile, the intra-regional differences in the southern marine economic circle increased, and the average annual growth rate of the intra-regional difference coefficient was 4.59%, indicating that the intra-regional differences were gradually widening in the development process. The marine industry development in coordination is to be improved in the future.

Columns (5) to (7) reveal the evolution of inter-regional differences in marine economic circles. The mean value of the inter-regional Gini coefficient showed significant inter-regional differences. From the evolutionary process, the inter-regional differences in the northern-eastern showed decreasing trends and its average annual decreasing rate was 3.10%. The inter-regional differences in northern-southern and eastern-southern showed inverted "U" types that increased initially and then decreased trends during the observation period.

Columns (8) to (10) report the contribution rates of different sources. From the average trend of the observation period, the annual average contribution rates of intra-regional, inter-regional, and intensity of transvariation were 30.5099%, 21.7047%, and 47.7854%, respectively. The contribution of the intensity of transvariation was the largest, indicating that the cross-over phenomenon of coastal provinces in the observation period was more significant. In other words, not all provinces in the marine economic circle with high MIDI had higher MIDI than provinces in other marine economic circles, so attention should be paid to the coordinated development of marine industry among provinces, especially the development of marine industry in the backward provinces in the economic circle with the high development level.

5. Conclusion and suggestions

5.1. Conclusion

For coastal countries, the ocean is not only a treasure trove of resources with great potential for development but also a key strategic space to support future development. Coastal countries are paying more and more attention to the development and utilization of marine resources and space, and the study of the marine industry development level has naturally become a focus of academic and political circles. In order to make up for the shortcomings of existing studies in scientifically evaluating the marine industry development level and analyzing its regional differences, this study constructs a comprehensive index system based on the marine industry development connotation and evaluates the overall and regional China's marine industry development level using the entropy-based TOPSIS method. Subsequently, the characteristics of regional differences are analyzed through the Dagum Gini coefficient. The findings are as follows: (1) The ranking of the evaluation system weights are innovation, outcomes, and industrial structure in order, with weights of 0.4597, 0.3943, and 0.1460, respectively, indicating that innovation can most reflect the marine industry development level. (2) The marine industry development level has been improving from the perspective of the whole country, the marine economic circle, and coastal provinces. And the progress is especially obvious after the strategy of building a strong maritime country is proposed, which indicates that national attention and policy support can effectively promote the development of the marine economy. (3) The intra-regional, inter-regional, and total differences in the other economic circles have decreased, except for the intra-regional differences in the southern marine economic circle, which indicates that the coordination of regional development has increased and marine policies have a significant role in promoting it. (4) Regional differences mainly originate from the intensity of transvariation, indicating that there are provinces with "abnormal values" in a marine economic circle, there are a small number of provinces whose MIDI does not align with the MIDI of the maritime economic circle in which they are located. So, attention should be paid to the coordinated development of marine industry among provinces, especially the backward provinces in the economic circle with high development level.

The conclusion of this paper is based on the existing data. Hence, with more sufficient data in the future, it is necessary to further recalculate the specific quantitative indicators. Besides, a possible next step of this study is to further explore the factors affecting the marine industry development and analyze the differentiation mechanism of its spatial and temporal evolution characteristics.

5.2. Suggestions

Based on the comprehensive index system and regional differences analysis of the marine industry development level in China, and to further release the potential of China's marine industry, the following suggestions are made focusing on innovation, industrial structure, and outcomes.

High-level marine scientific research platforms with shared resources and integrated capabilities should be established, which can promote effective integration of scientific research forces mainly from marine-related research institutions, universities, and enterprises. In addition, the innovation potential of ocean-related enterprises should be further stimulated, and communication and cooperation among research institutions, ocean-related enterprises, government, and other departments should be strengthened to establish a synergistic promotion mechanism and improve the transformation rate of scientific and technological achievements.

Optimize the layout and structure of the marine industry. The strategic marine emerging industries with high technology should be focused on cultivation, such as the new marine energy industry, marine information industry, deep-sea mining industry, and deep-sea aquaculture industry. At the same time, traditional marine industries with advantages such as marine fishery, marine transportation, and coastal tourism should be optimized and upgraded, and promoted in the direction of specialization, efficiency, and intelligence. Besides, the deep integration of marine industries and other industries should be actively encouraged, such as the integration of marine fishery and photovoltaic industry, and the integration of marine industry and digital information industry.

Additionally, persist in the concept of green development and promote harmonious coexistence between people and the ocean. Marine development and protection should be scientifically balanced, and market-oriented and sustainable paths for realizing the value of ecological products should be explored. It is necessary to establish a system of marine ecological protection and restoration and scientifically implement a series of major marine ecological protection and restoration projects, which can steadily improve the quality and stability of the marine ecosystem.

CRedit authorship contribution statement

Jianyue Ji: Conceptualization, Funding acquisition, Project administration, Supervision, Visualization, Writing – review & editing. **Huimin Liu:** Conceptualization, Data curation, Software, Methodology, Writing - original draft. **Xingmin Yin:** Validation, Software, Project administration, Resources, Formal analysis, Methodology, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The work was supported by the National Social Science Fund of China, Grant Number 19VHQ007.

References

- [1] R. Zhao, S. Hynes, G.S. He, Defining and quantifying China's ocean economy, *Mar. Pol.* 43 (2014) 164–173, <https://doi.org/10.1016/j.marpol.2013.05.008>.
- [2] P.M. Gilliland, D. Laffoley, Key elements and steps in the process of developing ecosystem-based marine spatial planning, *Mar. Pol.* 32 (2008) 787–796, <https://doi.org/10.1016/j.marpol.2008.03.022>.
- [3] E. Stebbings, E. Papathanasopoulou, T. Hooper, M.C. Austen, X.Y. Yan, The marine economy of the United Kingdom, *Mar. Pol.* 116 (2020), 103905, <https://doi.org/10.1016/j.marpol.2020.103905>.
- [4] Y. Sun, Y. Han, L.F. Zhang, Y.C. Chang, The construction and governance of regional maritime security order in the South China Sea: Conference report, *Mar. Pol.* 126 (2021), 104438, <https://doi.org/10.1016/j.marpol.2021.104438>.
- [5] J. Sun, J. Miao, J. Xu, N. Zhai, H. Mu, Sustainable development in marine economy: assessing carrying capacity of Shandong province in China, *OCRM* 216 (2022), 105981, <https://doi.org/10.1016/j.ocecoaman.2021.105981>.
- [6] M. Song, X.Y. Pan, X.F. Pan, Sustainable Marine Resource Utilization in China: A Comprehensive Evaluation, Elsevier, 2019 [http://refhub.elsevier.com/S0964-5691\(21\)00464-6/sref39](http://refhub.elsevier.com/S0964-5691(21)00464-6/sref39).
- [7] W.H. Ren, Y. Wang, J.Y. Ji, Research on China's marine economic growth pattern: An empirical analysis of China's eleven coastal regions, *Mar. Pol.* 87 (2018) 158–166, <https://doi.org/10.1016/j.marpol.2017.10.021>.
- [8] K.D. Yin, K. Zhang, C. Huang, Institutional supply, market cultivation, and the development of marine strategic emerging industries, *Mar. Pol.* 139 (2022), 105045, <https://doi.org/10.1016/j.marpol.2022.105045>.
- [9] Y.X. Wang, N. Wang, The role of the marine industry in China's national economy: an input–output analysis, *Mar. Pol.* 99 (2019) 42–49, <https://doi.org/10.1016/j.marpol.2018.10.019>.
- [10] X. Yan, L. Yan, X.L. Yao, M. Liao, The marine industrial competitiveness of blue economic regions in China, *Mar. Pol.* 62 (2015) 153–160, <https://doi.org/10.1016/j.marpol.2015.09.015>.
- [11] H.D. Ma, L.X. Li, Could environmental regulation promote the technological innovation of China's emerging marine enterprises? Based on the moderating effect of government grants, *Environ. Res.* 202 (2021), 111682, <https://doi.org/10.1016/j.envres.2021.111682>.
- [12] B.S. Halpern, C. Longo, D. Hardy, K.L. McLeod, J.F. Samhouri, S.K. Katona, D. Zeller, An index to assess the health and benefits of the global ocean, *Nature* 488 (2012) 615–620, <https://doi.org/10.1038/nature11397>.
- [13] Y. Zhang, S.H. Wang, Influence of marine industrial agglomeration and environmental regulation on marine innovation efficiency—From an innovation value chain perspective, *Mar. Pol.* 134 (2021), 104807, <https://doi.org/10.1016/j.marpol.2021.104807>.
- [14] Q.L. Shao, L.J. Chen, R.Y. Zhong, H.T. Weng, Marine economic growth, technological innovation, and industrial upgrading: a vector error correction model for China, *Ocean Coast. Manag.* 200 (2021), 105481, <https://doi.org/10.1016/j.ocecoaman.2020.105481>.
- [15] L.L. Wang, M. Su, H. Kong, Y.X. Ma, The impact of marine technological innovation on the upgrade of China's marine industrial structure, *Ocean Coast. Manag.* 211 (2021), 105792, <https://doi.org/10.1016/j.ocecoaman.2021.105792>.
- [16] S. Xu, B.B. Lu, Q.D. Yue, Impact of sci-tech finance on the innovation efficiency of China's marine industry, *Mar. Pol.* 133 (2021), 104708, <https://doi.org/10.1016/j.marpol.2021.104708>.
- [17] B.S. Halpern, C. Longo, C. Scarborough, D. Hardy, B.D. Best, S.C. Doney, J. F. Samhouri, Assessing the health of the US West coast with a regional-scale application of the ocean health index, *Plos One* 9 (2014), e98995, <https://doi.org/10.1371/journal.pone.0098995>.
- [18] B.S. Halpern, Building on a decade of the ocean health index, *One Earth* 2 (2020) 30–33, <https://doi.org/10.1016/j.oneear.2019.12.011>.
- [19] J.L. Anderson, C.M. Anderson, J. Chu, J. Meredith, F. Asche, G. Sylvia, D. Valderrama, The fishery performance indicators: a management tool for triple bottom line outcomes, *Plos One* 10 (2015), e0122809, <https://doi.org/10.1371/journal.pone.0122809>.
- [20] W.L. Song, J. Guo, Y. Wang, L.L. Li, An evaluation of the development of the ocean economy in China using an ocean economic development index, *Mar. Pol.* 132 (2021), 104691, <https://doi.org/10.1016/j.marpol.2021.104691>.
- [21] F.Z. Li, W.X. Xing, M. Su, J. Xu, The evolution of China's marine economic policy and the labor productivity growth momentum of marine economy and its three economic industries, *Mar. Pol.* 134 (2021), 104777, <https://doi.org/10.1016/j.marpol.2021.104777>.
- [22] J.Y. Ji, D.F. Wang, Y. Zhang, Marine biopharmaceutical industrial policies in China: evolution, characteristics, and challenges, *Ocean Coast. Manag.* 225 (2022), 106224, <https://doi.org/10.1016/j.ocecoaman.2022.106224>.
- [23] S.S. Jiang, J.M. Li, Do political promotion incentive and fiscal incentive of local governments matter for the marine environmental pollution? Evidence from China's coastal areas, *Mar. Pol.* 128 (2021), 104505, <https://doi.org/10.1016/j.marpol.2021.104505>.
- [24] J.A. Schumpeter, *The theory of economic development: an inquiry into profits, capital, credit, interest, and the business cycle*, *Harv. Econ. Stud.* 46 (1949).
- [25] J.Y. Ji, X.W. Sun, Research on the connotation and evaluation framework of the transformation and upgrading of marine industry, *J. Ocean U. China (Soc. Sci. Ed.)* 6 (2021) 33–40, <https://doi.org/10.16497/j.cnki.1672-335X.202106004>.
- [26] P.D. Liu, B.Y. Zhu, M.Y. Yang, Has marine technology innovation promoted the high-quality development of the marine economy? —Evidence from coastal regions in China, *Ocean Coast. Manag.* 209 (2021), 105695, <https://doi.org/10.1016/j.ocecoaman.2021.105695>.

- [27] J. Li, B. Yu, Strategic or substantive innovation?—The impact of institutional investors' site visits on green innovation evidence from China, *Technol. Soc.* 68 (2022), 101904, <https://doi.org/10.1016/j.techsoc.2022.101904>.
- [28] Y. Su, Q.M. Fan, Renewable energy technology innovation, industrial structure upgrading and green development from the perspective of China's provinces, *Technol. Forecast. Soc. Change* 180 (2022), 121727, <https://doi.org/10.1016/j.techfore.2022.121727>.
- [29] R.M. Auty, M. Richard, Pollution patterns during the industrial transition, *Geogr. J.* 163 (1997) 206–215, <https://doi.org/10.2307/3060184>.
- [30] H. Theil, Economics and information theory, *Econ. Rev.* 19 (1967) 185–188.
- [31] H.B. Chenery, H. Elkington, C.A. Sims, *A Uniform Analysis Of Development Patterns*, Harvard University, Center for International Affairs, 1970.
- [32] P. Krugman, Increasing returns and economic geography, *J. Polit. Econ.* 99 (1991) 483–499, <https://doi.org/10.1086/261763>.
- [33] Z. Tang, An integrated approach to evaluating the coupling coordination between tourism and the environment, *Tour. Manag* 46 (2015) 11–19, <https://doi.org/10.1016/j.tourman.2014.06.001>.
- [34] J.H. Moore, A measure of structural change in output, *Rev. Income Wealth* 24 (1987) 105–118, <https://doi.org/10.1111/j.1475-4991.1978.tb00034.x>.
- [35] Z.H. Cheng, L.S. Li, J. Liu, Industrial structure, technical progress and carbon intensity in China's provinces, *Renew. Sustain. Energy Rev.* 81 (2017) 2935–2946, <https://doi.org/10.1016/j.rser.2017.06.103> pt.2.
- [36] C.G. Liu, W. Sun, P.X. Li, Characteristics of spatiotemporal variations in coupling coordination between integrated carbon emission and sequestration index: a case study of the Yangtze River Delta, China, *Ecol. Indic.* 135 (2022), 108520, <https://doi.org/10.1016/j.ecolind.2021.108520>.
- [37] H. Schandl, S. Hatfield-Dodds, T. Wiedmann, Decoupling global environmental pressure and economic growth: scenarios for energy use, materials use and carbon emissions, *J. Clean. Prod.* 132 (2016) 45–56, <https://doi.org/10.1016/j.jclepro.2015.06.100>.
- [38] B. Yin, Y. Pang, X. Xu, The path of housing prices in promoting the upgrading of industrial structure: bank credit funds, land finance, and consumer demand, *Sustain* 14 (2022) 5731, <https://doi.org/10.3390/su14095731>.
- [39] V. Salehi, H. Zarei, Gh.A. Shirali, K. Hajizadeh, An entropy-based TOPSIS approach for analyzing and assessing crisis management systems in petrochemical industries, *J. Loss Prev. Process Ind.* 67 (2022), 104241, <https://doi.org/10.1016/j.jlp.2020.104241>.
- [40] X.H. He, Q.Y. Ping, W.F. Hu, Does digital technology promote the sustainable development of the marine equipment manufacturing industry in China? *Mar. Pol.* 136 (2022), 104868 <https://doi.org/10.1016/j.marpol.2021.104868>.
- [41] X. Liu, X. Zhou, B. Zhu, K. He, P. Wang, Measuring the maturity of carbon market in China: An entropy-based TOPSIS approach, *J. Clean. Prod.* 229 (2019) 94–103, <https://doi.org/10.1016/j.techfore.2019.119874>.
- [42] M. Zhou, X.B. Liu, J.B. Yang, Y.W. Chen, J. Wu, Evidential reasoning approach with multiple kinds of attributes and entropy-based weight assignment, *Knowle-Based Syst.* 163 (2019) 358–375, <https://doi.org/10.1016/j.knosys.2018.08.037>.
- [43] C. Dagum, A new approach to the decomposition of the Gini income inequality ratio, *Empir. Econ.* 22 (1997) 515–531, <https://doi.org/10.1007/BF01205777>.